ALEXANDER DOMBOWSKY, PH.D.

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HIGHLIGHT

Bioinformatics Fellow at Gladstone Institutes and Stanford with a PhD in Statistics from Duke. Research focuses on developing principled AI and ML approaches to extract insights from complex data. Extensive experience in interdisciplinary collaborations, with multiple first-author publications.

RESEARCH INTERESTS

Probabilistic machine learning, AI for science, precision medicine, statistics, deep learning, computational biology, biomarker discovery, transcriptomics, proteomics, drug discovery, game-theoretic statistics, uncertainty quantification, clustering, graphs, multimodal data integration

EXPERIENCE

Stanford University & Gladstone Institutes — San Francisco Bay Area, CA, USA
Bioinformatics Fellow (Engelhardt Lab) — *Aug. 2025 - Present*

- Led ML methods development for biomarker discovery from transcriptomic data; wrote reproducible Python code for model selection, training, and interpretation
- Worked closely with biologists for data cleaning and biological interpretation, gave presentations in joint meetings
- Designed a sequential test for ML models that decreased detection time by 48% on average compared to standard approach and achieved approximate theoretical optimality
- Led "Deep Learning 4 Science" reading group at Gladstone, gave lectures on deep neural networks, transformers, and protein structure prediction

Duke University — Durham, NC, USA
Research Assistant — *Aug. 2021 - July 2025*

- Developed Bayesian clustering algorithm that achieved an 8% increase in accuracy for cell-type recovery over state-of-the-art methods; created associated R package with detailed documentation
- Led methods development in collaboration with clinicians that found phenotypes of sepsis patients with distinct medical interpretations using a novel Bayesian clustering prior
- Created Bayesian multimodal clustering and graph feature selection methods, applied them to epidemiological and breast cancer datasets to uncover latent clustering and network structure
- Mentored junior PhD students, organized reading groups for computational methods and modeling environmental exposures

McGill University — Montreal, QC, Canada
Research Assistant — *Sept. 2019 - July 2020*

- Demonstrated computational pathologies from Hamiltonian Monte Carlo through case studies, recommended solutions for improving inferences
- Led an independent study course on Bayesian optimization, implemented key algorithms in Python and applied them to standard benchmark examples

EDUCATION

Duke University — Durham, NC, USA
Doctor of Philosophy – Statistical Science — *Aug. 2020 - May 2025*

- Advisors: David B. Dunson and Amy H. Herring
- Dissertation: *Bayesian Inference for Discrete Structures*
- Myra and William Waldo Boone Fellowship — *2023-2025*

McGill University Montreal, QC, Canada
Master of Science – Mathematics and Statistics *Sept. 2019 - July 2020*

- Advisor: Russell Steele

McGill University Montreal, QC, Canada
Bachelor of Science – Mathematics *Sept. 2015 - May 2019*

PUBLICATIONS

Dombowsky, A., Dunson, D. B., Madut, D. B., Rubach, M. P., & Herring, A. H. (2025). Bayesian learning of clinically meaningful sepsis phenotypes in northern Tanzania. *The Annals of Applied Statistics.* [Publication] [Preprint] [Code].

Dombowsky, A. & Dunson, D. B. (2025). Product centered Dirichlet processes for Bayesian multiview clustering. *Journal of the Royal Statistical Society, Series B.* [Publication] [Preprint] [Code].

Dombowsky, A., & Dunson, D. B. (2025). Bayesian clustering via fusing of localized densities. *Journal of the American Statistical Association: Theory and Methods.* [Publication] [Preprint] [Package].

SUBMITTED PAPERS

Dombowsky, A., Engelhardt, B. E., & Ramdas, A. (2026). Besag-Clifford e-values for testing an unnormalized null. [Preprint] [Code].

Dombowsky, A. & Dunson, D. B. (2025). Learning discrete Bayesian networks with hierarchical Dirichlet shrinkage. [Preprint] [Code].

Madut, D. B., **Dombowsky, A.** [& 16 others]. (2025). Derivation of clinical sepsis subtypes in northern Tanzania using Bayesian probabilistic clustering of clinical data.

PAPERS IN PREPARATION

Meng, J., **Dombowsky, A.**, & Dunson, D. B. (2026). Kernel robust Bayesian clustering in high dimensions.

HONORS AND AWARDS

ASA Statistics in Epidemiology Early Career Award *Aug. 2026*

- Among four awardees of PhD students and postdocs in the ASA
- Provided funding to present at JSM 2026

Myra and William Waldo Boone Fellowship for Canadian Graduate Students *2023-2025*

- Among two or three awardees in the 2023-24 and 2024-25 academic years

ASA Bayesian Statistics Student Paper Competition *Aug. 2024*

- Among ten awardees of PhD students and postdocs in the ASA
- Provided funding to present at JSM 2024

ISBA World Meeting Travel Award *2022, 2024*

BAYSM 2023 Best Long Talk Award *Nov. 2023*

SKILLS

Programming & Tools: R (base, dplyr, ggplot2, Bioconductor), Python (NumPy, scikit-learn, pandas; familiar with PyTorch and JAX), C++ (via Rcpp), Linux/Unix, Git, MATLAB

Machine Learning: Supervised & unsupervised learning (regression, classification, clustering, dimensionality reduction), feature selection, model evaluation, deep neural networks, transformers, conformal prediction

Statistics: Probabilistic modeling, Bayesian inference, uncertainty quantification, hypothesis testing, p-values and e-values, probabilistic graphical models, hierarchical models, robustness, time series & stochastic processes

Other: Reproducible code, scientific communication

TALKS

Invited talk, "Besag-Clifford e-values for unnormalized testing", SSC and CMStatistics, 2026

Invited talk, "Product centered Dirichlet processes for dependent clustering", ISBA, 2024

Contributed talk, "Learning discrete Bayesian networks with hierarchical Dirichlet shrinkage", JSM, 2026–SIE Early Career Award

Contributed talk, "Bayesian conformal prediction via score modeling", ISBA, 2026

Contributed talk, "Hierarchical directed Dirichlet networks for discrete graphical modeling", BNP and BAYSM, 2025

Contributed talk, "Product centered Dirichlet processes for dependent clustering", JSM, 2024–SBSS Student Paper Award

Contributed talk, "Bayesian clustering via fusing of localized densities", BAYSM, 2023–Best Long Talk Award

TEACHING EXPERIENCE

Teaching Assistant at Duke University *Jan. 2021 - Dec. 2024*

- STA 490/690: Analysis of Time-to-Event Data (Instructor: Yue Jiang)
- STA 831: Probability & Statistical Models (Instructor: Mike West)
- STA 470S: Introduction to Statistical Consulting (Instructor: Edwin Iversen)
- STA 360: Bayesian Methods and Modern Statistics (Instructor: Simon Mak)

Teaching Assistant at McGill University *Sept. 2019 - Dec. 2019*

- MATH 203: Principles of Statistics 1 (Instructors: Abbas Khalili and David Wolfson)

PROFESSIONAL SERVICES AND ACTIVITIES

Reviewer: Journal of the American Statistical Association (JASA), Electronic Journal of Statistics (EJS), Biometrika, ASA SBSS Student Paper Competition

Member: ISBA, j-ISBA, the ASA (Biometrics & SBSS), and the IBS (ENAR)

Session chair: BAYSM 2024–"Bayesian methods for environmental and health data"